

Proposal for the Development of a National Horticulture Crop Tracking & Border Policy Management System

1. INTRODUCTION

This brief presents a proposal for the design and development of a digital system to support the Ministry of Agriculture's Horticulture Division in the collection, analysis and application of real-time horticultural crop planting data from registered horticulture farmers across all districts of Botswana. The system is intended to enhance the Ministry's capacity to monitor national food (horticultural produce) supply, generate timely situational reports and provide evidence-based recommendations on the management of horticulture produce imports and exports.

2. PROBLEM STATEMENT

At present, the Horticulture Division lacks a reliable, real-time centralised mechanism for tracking what farmers are planting, how much land is under cultivation and what quantities of produce are expected at harvest. This information gap has two direct and costly consequences for the country:

1. When produce is in short supply, the shortage is often identified too late for the Ministry to take corrective action in time, resulting in price spikes, food insecurity and reactive [rather than planned] import decisions.
2. When farmers produce a bumper harvest and borders remain open to cheaper imported produce, local farmers suffer significant income losses because they cannot compete on price, undermining the long-term viability of Botswana's domestic horticulture sector.

The system is proposed as a direct solution to both challenges by giving the Ministry advance visibility, weeks or months before harvest, into what the national supply picture will look like.

3. PROPOSED SYSTEM OVERVIEW

The system will operate across three integrated portals, each tailored to a specific category of user within the horticulture value chain:

Farmer Registration Portal

- ▶ Login secured via National ID (Oman)
- ▶ Submit crop type, hectares, planting date and irrigation method
- ▶ Receive automatic yield estimate on submission
- ▶ SMS confirmation with unique reference number
- ▶ USSD access (*short code*) for farmers without smartphones
- ▶ View full history of past crop declarations

District Agric Officer & Department of Crop Production (Boarder Policy Advisory)

- ▶ Real-time national and district-level crop overviews
- ▶ Automated supply vs. demand comparison per crop
- ▶ Deficit and surplus alerts sent to relevant officers
- ▶ Border policy recommendations updated every 24 hours
- ▶ Exportable situational reports in PDF and Excel
- ▶ Role-based access: District, National and Ministerial levels

4. AUTOMATED YIELD ESTIMATION

Upon each farmer submission, the system will automatically calculate the expected harvest yield using a configurable formula that accounts for the crop type, area planted, irrigation method and the planting season. No manual calculation will be required from either the farmer or the officer. These estimates will

feed directly into national projection totals visible on the officer dashboard in real time.

Illustration: A farmer in Kweneng District submits that they have planted 12.5 hectares of tomatoes using drip irrigation during the main season. The system immediately calculates an estimated yield of **431 tonnes** and adds this figure to the national tomato projection total, updating all officer dashboards in real time.

Formula: Expected Yield (tonnes) = Hectares × Base Yield (t/ha)

5. BORDER POLICY DECISION SUPPORT

The system will include a dedicated Border Policy Advisor module for use by senior officials. This module will continuously compare projected national supply against Botswana's estimated national demand, sourced from Statistics Botswana consumption figures or the Department of Crop Production and generate automatic recommendations as follows:

Projected Supply Level	Recommendation	Rationale
110% or more of demand	Restrict Imports	Domestic surplus is sufficient. Open imports would undermine local farmer income.
75% to 109% of demand	Monitor Supply	Borderline situation. Weekly review recommended with contingency imports on standby.
Below 75% of demand	Open Borders	Supply deficit is likely. Immediate import authorisation advised to prevent shortages.

Officials (Department of Crop Production @ MoA) retain full authority to accept, modify or override any recommendation. All decisions and their justifications will be logged within the system for accountability and audit purposes.

6. PROPOSED IMPLEMENTATION PLAN

PHASE 1 Pilot Months 1–4 Two districts (Kweneng, South East). Selected crops (tomatoes, onions). Farmer registration and officer dashboard activated.	PHASE 2 National Rollout Months 5–8 All districts. All tracked crops. Mobile application and USSD service activated nationwide.	PHASE 3 Policy Engine Months 9–12 Integration with Botswana Met Services and BURS. Border Advisor module live. Automated deficit/surplus alerts active.	PHASE 4 AI Forecasting Year 2 onward Predictive models trained on two to three seasons of historical data for long-range national supply projections.
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7. DATA GOVERNANCE & LEGAL COMPLIANCE

The system will be developed in full compliance with Botswana's **Data Protection Act (2018)**. Farmer personal data will be accessible only to authorised personnel within their respective districts. Officers at all levels will access individual and aggregated farmer records. A full audit trail will be maintained for all data entries, edits and policy decisions made within the system. Data shall be hosted within Botswana compliant cloud infrastructure.

8. RECOMMENDATION

It is respectfully recommended that the Ministry of Agriculture's Horticulture Division gives favourable

consideration to the commissioning of this system as a priority digital infrastructure investment. The system directly addresses the Ministry's mandate to ensure national food security, support the livelihoods of horticulture farmers and enable informed, evidence-based trade policy decisions on behalf of the Republic of Botswana.

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